

Ensemble Size Optimization for CMIP6 Rainfall Downscaling in Coastal Thailand: A Bootstrap-Validated Saturation Approach

การกำหนดขนาดกลุ่มแบบจำลอง **CMIP6** ที่เหมาะสมสำหรับการปรับรายละเอียดข้อมูลฝนในพื้นที่ชายฝั่งประเทศไทย ด้วยวิธีจุดอิ่มตัวที่ตรวจสอบด้วยบูตสเตรป

Abstract

ความน่าเชื่อถือของการประเมินผลกระทบจากการเปลี่ยนแปลงสภาพภูมิอากาศในพื้นที่ชายฝั่งเขตร้อนไม่ได้ขึ้นอยู่กับคุณภาพของแบบจำลองเพียงอย่างเดียว แต่ยังขึ้นอยู่กับข้อกำหนดขนาดกลุ่มแบบจำลองที่เหมาะสมระหว่างความแม่นยำของการจำลองกับต้นทุนด้านการคำนวณ ภายใต้สภาพภูมิอากาศที่ได้รับอิทธิพลจากมรสุม การเพิ่มจำนวนแบบจำลองมากเกินไปจนอาจเพิ่มภาระการประมวลผลโดยไม่ก่อให้เกิดการปรับปรุงผลการจำลองอย่างมีนัยสำคัญ งานวิจัยนี้พัฒนากรอบการวิเคราะห์จุดอิ่มตัวของกลุ่มแบบจำลอง (**ensemble saturation framework**) ร่วมกับการตรวจสอบความเชื่อมั่นด้วยวิธีบูตสเตรป เพื่อกำหนดจำนวนแบบจำลอง **CMIP6** ที่เหมาะสมสำหรับการปรับรายละเอียดข้อมูลฝนรายวันในจังหวัดประจวบคีรีขันธ์ ซึ่งเป็นพื้นที่ชายฝั่งภาคใต้ของประเทศไทยที่มีพลวัตฝนมรสุมซับซ้อน โดยใช้ข้อมูลฝนรายวันจากสถานีอุตุนิยมวิทยา 12 สถานีในช่วงปี พ.ศ. 2524–2557 เปรียบเทียบกับทุกชุดความเป็นไปได้ของแบบจำลอง **CMIP6** จำนวน 6 แบบจำลอง ประสิทธิภาพของแบบจำลองได้รับการประเมินทั้งก่อนและหลังการปรับแก้ด้วยวิธี **Quantile Delta Mapping (QDM)** ผ่านดัชนี **RMSE**, **NSE**, **KGE** และค่าสหสัมพันธ์เพียร์สัน (r) ขณะที่จุดอิ่มตัวของกลุ่มแบบจำลองถูกระบุด้วยเกณฑ์การลดลงของผลตอบแทนส่วนเพิ่มและตรวจสอบด้วยการสุ่มตัวอย่างซ้ำแบบบูตสเตรปจำนวน 999 ครั้ง ผลการศึกษาพบว่าแบบจำลอง **CMIP6** ดิบมีอคติเชิงระบบสูงและให้ค่า **NSE** ติดลบในทุกขนาดกลุ่มแบบจำลอง สะท้อนข้อจำกัดในการจำลองความแปรปรวนของฝนที่สังเกตได้ การปรับแก้ด้วย **QDM** ช่วยยกระดับประสิทธิภาพของทุกดัชนีและลดความไม่แน่นอนระหว่างชุดแบบจำลองย่อยอย่างชัดเจน การวิเคราะห์จุดอิ่มตัวชี้ว่า **RMSE** และ **NSE** เข้าสู่ภาวะอิ่มตัวที่จำนวน 5 แบบจำลอง ขณะที่ **KGE** และ r เข้าสู่ภาวะอิ่มตัวที่ 6 แบบจำลอง โดยผลการตรวจสอบด้วยบูตสเตรปยืนยันความมีเสถียรภาพของค่าดังกล่าวและแสดงความแปรปรวนในการสุ่มตัวอย่างที่ต่ำมาก หลังจากจำนวนแบบจำลองเกิน 5 แบบจำลอง การเพิ่มประสิทธิภาพมีน้อยมาก โดย **RMSE** ดีขึ้นเพียง 1.1% และแทบไม่พบการปรับปรุงเพิ่มเติมของ **KGE** และ r การวิเคราะห์เชิงพื้นที่ยังพบว่าจุดอิ่มตัวของ **RMSE** และ **NSE** มีความสอดคล้องกันทุกสถานี ขณะที่ **KGE** แสดงความแตกต่างเฉพาะพื้นที่ โดยมีสองสถานีที่เข้าสู่ภาวะอิ่มตัวเร็วกว่าสถานีอื่นเนื่องจากมีทักษะของแบบจำลองสูงกว่าและมีลักษณะการกระจายของฝนที่ซับซ้อนน้อยกว่า ผลการศึกษาชี้ว่ากลุ่มแบบจำลอง **CMIP6** ที่ผ่านการปรับแก้จำนวน 5 แบบจำลองสามารถรักษาศักยภาพการจำลองฝนได้เกือบทั้งหมดในสภาพแวดล้อมชายฝั่งที่ได้รับอิทธิพลจากมรสุม กรอบวิธีการที่นำเสนอสามารถประยุกต์ใช้เป็นแนวทางเชิงระบบสำหรับการกำหนดขนาดกลุ่มแบบจำลองที่เหมาะสม สนับสนุนการปรับรายละเอียดข้อมูลภูมิอากาศ การจำลองทางอุทกวิทยา และการวางแผนปรับตัวต่อการเปลี่ยนแปลงสภาพภูมิอากาศในพื้นที่ชายฝั่งเขตร้อนที่มีลักษณะใกล้เคียงกัน

The reliability of climate impact assessments in tropical coastal regions depends not only on model quality but also on selecting an ensemble size that balances simulation skill against computational cost. In monsoon-dominated environments, unnecessarily large ensembles may increase processing demands without yielding meaningful improvements in predictive performance. This study developed a bootstrap-validated saturation framework to determine the optimal number of CMIP6 models required for daily rainfall downscaling in Prachuap Khiri Khan Province, a coastal region of southern Thailand characterized by complex monsoonal rainfall dynamics. Daily precipitation records from 12 meteorological stations during 1981–2014 were evaluated against all possible combinations of six CMIP6 models. Model performance was assessed before and after Quantile Delta Mapping (QDM) bias correction using root mean square error (RMSE), Nash–Sutcliffe efficiency (NSE), Kling–Gupta efficiency (KGE), and Pearson correlation coefficient (r). Ensemble saturation thresholds were identified using a marginal-gain criterion and verified through 999 bootstrap resampling experiments. Raw CMIP6 simulations exhibited substantial systematic biases, with negative NSE values across all ensemble sizes, indicating limited ability to reproduce observed rainfall variability. QDM markedly improved performance and reduced inter-combination uncertainty, yielding consistent gains in RMSE, NSE, KGE, and correlation. Saturation analysis revealed a stable threshold at five ensemble members for RMSE and NSE, while KGE and r reached saturation at six members. Bootstrap confidence intervals confirmed the robustness of these thresholds, with negligible sampling variability. Beyond five models, additional skill gains were operationally insignificant, amounting to only 1.1% for RMSE and effectively zero for KGE and correlation. Spatial diagnostics showed uniform RMSE and NSE saturation across all stations, whereas KGE saturation exhibited localized heterogeneity, with two stations achieving convergence at smaller ensemble sizes due to higher baseline model skill and reduced precipitation complexity. The results demonstrate that a five-member bias-corrected CMIP6 ensemble captures nearly all attainable rainfall simulation skill in this coastal monsoonal setting. The proposed saturation framework provides a reproducible approach for ensemble optimization and offers practical guidance for computationally efficient climate downscaling, hydrological modelling, and climate adaptation planning across comparable tropical coastal regions.

ทางเลือก

บทคัดย่อ (Abstract)

งานวิจัยนี้มุ่งตอบคำถามสำคัญในทางปฏิบัติและระเบียบวิธีสำหรับการจำลองสภาพภูมิอากาศระดับภูมิภาค (Regional climate downscaling) ว่าต้องใช้แบบจำลอง CMIP6 จำนวนเท่าใดจึงจะเพียงพอต่อการจำลองปริมาณน้ำฝนรายวันให้มีประสิทธิภาพคงที่และใกล้เคียงระดับสูงสุด เมื่อใช้วิธีการปรับแก้ความคลาดเคลื่อนแบบ Quantile Delta Mapping (QDM) การศึกษาได้ดำเนินการวิเคราะห์แบบจัดหมู่กลุ่มแบบจำลองย่อยแบบครบถ้วน (Exhaustive combinatorial sub-ensemble analysis) ในทุกรูปแบบความเป็นไปได้ ณ สถานีตรวจวัด 12 แห่งในจังหวัดประจวบคีรีขันธ์ และได้สร้างกรอบแนวคิดจุดอิ่มตัวที่ตรวจสอบความถูกต้องด้วยเทคนิคบูตสแตรป (Bootstrap-validated saturation framework) เพื่อระบุขนาดกลุ่มแบบจำลองขั้นต่ำที่จุดซึ่งการเพิ่มจำนวนแบบจำลองไม่ส่งผลต่อการเพิ่มประสิทธิภาพอย่างมีนัยสำคัญในทางปฏิบัติ

ผลการศึกษาหลักพบว่า กลุ่มแบบจำลอง CMIP6 ที่ผ่านการปรับแก้ด้วยวิธี QDM จำนวน 5 แบบจำลอง ถือเป็นขนาดที่สามารถให้ประสิทธิภาพการจำลองปริมาณน้ำฝนรายวันที่ครอบคลุมได้ดีที่สุดในพื้นที่ชายฝั่งมรสุมแห่งนี้ การเพิ่มแบบจำลองที่ 6 เข้าไปให้ผลลัพธ์ที่ดีขึ้นเพียงเล็กน้อย (ค่า RMSE ลดลงสูงสุดเพียง 1.1% และไม่พบการเพิ่มขึ้นของค่า KGE และค่าสหสัมพันธ์เพียร์สัน) จุดอิมิตัวนี้ได้รับการยืนยันผ่านการจำลองสุ่มซ้ำด้วยวิธีบูตสเตรป (999 รอบ) โดยดัชนีชี้วัดที่ถ่วงน้ำหนักด้วยค่าความคลาดเคลื่อน (RMSE และ NSE) ถึงจุดอิมิตัวเร็วกว่าที่จำนวน 5 แบบจำลอง เนื่องจากการหาค่าเฉลี่ยกลุ่มแบบจำลองช่วยลดความคลาดเคลื่อนขนาดใหญ่ได้อย่างมีประสิทธิภาพ ขณะที่ดัชนีชี้วัดแบบพหุองค์ประกอบ (KGE และ r) ต้องการแบบจำลองครบทั้ง 6 ชุดเพื่อให้เกิดความสมดุลสูงสุด นอกจากนี้ วิธี QDM ยังช่วยแก้ปัญหการจำลองของ CMIP6 ดิบที่ไม่สามารถจำลองการกระจายตัวของฝนได้อย่างมีนัยสำคัญ

ในแง่ของความแปรปรวนเชิงพื้นที่ พบว่าจุดอิมิตัวของ RMSE และ NSE มีความสม่ำเสมอที่ 5 แบบจำลองในทุกสถานี แต่จุดอิมิตัวของ KGE มีความแตกต่างกัน โดยสถานีที่มีประสิทธิภาพการจำลองสูงสุดสามารถถึงจุดอิมิตัวได้เร็วกว่า (ที่ 3-4 แบบจำลอง) สะท้อนให้เห็นว่าความแปรปรวนเชิงพื้นที่ของปริมาณน้ำฝนมีผลต่อการออกแบบกลุ่มแบบจำลอง

แม้การศึกษานี้จะมีข้อจำกัดภายใต้กลุ่มแบบจำลอง 6 ชุดและระเบียบวิธีที่เลือกใช้ แต่ผลลัพธ์ที่ได้มีนัยสำคัญอย่างยิ่งต่อการดำเนินงานจริง สำหรับนักวางแผนทรัพยากรน้ำและนักอุทกวิทยาที่ต้องทำงานภายใต้ข้อจำกัดด้านทรัพยากรการประมวลผล การใช้แบบจำลอง CMIP6 ที่ผ่านการปรับแก้เพียง 5 ชุดสามารถให้ประสิทธิภาพเทียบเท่าการใช้ 6 ชุด กรอบการประเมินนี้สามารถนำไปประยุกต์ใช้กับพื้นที่ชายฝั่งมรสุมอื่น ๆ ในภูมิภาคเอเชียตะวันออกเฉียงใต้ เพื่อเป็นแนวทางในการปรับขนาดกลุ่มแบบจำลองทางอุทกภูมิอากาศให้มีประสิทธิภาพเชิงการคำนวณสูงสุดและมีความน่าเชื่อถือทางวิทยาศาสตร์

1. Introduction

Reliable rainfall projections are fundamental to climate change adaptation, water resources management, agricultural planning, flood mitigation, and drought preparedness. In tropical and monsoon-dominated regions, precipitation variability represents one of the most uncertain components of the climate system owing to the complex interactions among large-scale circulation, mesoscale convection, coastal processes, and local topographic effects. These challenges are particularly pronounced in Southeast Asia, where rainfall regimes are strongly influenced by seasonal monsoons and exhibit substantial spatial and temporal variability. Consequently, robust climate information at regional and local scales remains essential for supporting evidence-based decision-making under a changing climate.

The Coupled Model Intercomparison Project Phase 6 (CMIP6) provides the most advanced generation of global climate model simulations currently available for climate change assessment (Desmet, Q., & Ngo-Duc, T. (2021)). Compared with previous model generations, CMIP6 incorporates improvements in physical parameterizations, Earth system processes, aerosol forcing representation, and future socioeconomic pathways. As a result, CMIP6 simulations have become a primary source of climate information for impact assessment studies worldwide. Nevertheless, substantial uncertainties remain in precipitation projections because rainfall processes are governed by highly nonlinear atmospheric mechanisms that are often inadequately resolved at the spatial scales relevant to local hydrological applications.

Consequently, raw climate model outputs frequently exhibit systematic biases in precipitation magnitude, frequency, and distribution, particularly in tropical coastal environments.

To address these limitations, climate impact studies commonly employ multi-model ensembles rather than relying on individual climate models. The underlying rationale is that averaging across multiple models can reduce random model-specific errors and provide more robust estimates of future climate conditions. Multi-model ensembles have therefore become a standard framework in climate projection studies and have been widely adopted in successive assessment reports of the Intergovernmental Panel on Climate Change (IPCC). However, despite their widespread use, the interpretation and design of climate model ensembles remain challenging because climate models are not independent realizations of a common probability distribution and often share structural assumptions, parameterizations, and development histories (Ona et al., 2024). Consequently, increasing the number of models does not necessarily produce proportional improvements in simulation skill. Instead, the benefits of adding additional models may diminish beyond a certain threshold, creating a practical trade-off between ensemble size, computational cost, and predictive performance Asha, S., et al. (2024).

An additional challenge arises from the systematic biases inherent in global climate model simulations. Statistical bias-correction techniques have therefore become an essential component of climate downscaling workflows. Among the available approaches, Quantile Delta Mapping (QDM) has received increasing attention because it corrects distributional biases while explicitly preserving projected changes in precipitation quantiles Chen, X., et al. (2022); Supharatid et al., 2022). This characteristic is particularly important for hydrological applications where both mean conditions and extreme events contribute to risk assessment. Previous studies have demonstrated that QDM effectively improves precipitation simulations while maintaining climate change signals embedded within the original model outputs, making it one of the most widely adopted bias-correction approaches in contemporary climate impact research.

Although considerable effort has been devoted to evaluating climate model performance and developing bias-correction methodologies, comparatively little attention has been given to determining the optimal number of climate models required within a multi-model ensemble. Most ensemble-based studies employ all available models or select a subset based on subjective criteria, often without explicitly assessing whether additional models contribute meaningful predictive information. As a result, ensemble size is frequently determined by data availability rather than by quantitative evidence. This issue is particularly relevant for regional downscaling applications because computational demands increase rapidly with ensemble size, while the marginal gains in performance may become negligible beyond a certain point. Identifying an objective saturation threshold for ensemble performance could therefore improve computational efficiency while maintaining scientific rigor.

Coastal regions of Thailand provide an important setting for investigating this question. Prachuap Khiri Khan Province, located along the western coast of the Gulf of Thailand, experiences pronounced monsoonal rainfall variability and recurrent hydroclimatic hazards, including seasonal flooding, water shortages, and extreme precipitation events. Accurate rainfall simulations are therefore essential for climate adaptation planning and hydrological impact assessments in the region. However, the extent to which increasing the number of CMIP6 models improves rainfall simulation skill after bias correction remains largely unknown.

Against this background, the present study develops a bootstrap-validated ensemble saturation framework to quantify the relationship between ensemble size and rainfall simulation performance in a tropical coastal environment. Daily rainfall observations from twelve meteorological stations during 1981–2014 were compared against all possible sub-ensemble combinations derived from six CMIP6 models before and after Quantile Delta Mapping bias correction. Ensemble performance was evaluated using root mean square error (RMSE), Nash–Sutcliffe efficiency (NSE), Kling–Gupta efficiency (KGE), and Pearson correlation coefficient (r), while uncertainty reduction and spatial consistency were assessed across the station network. A bootstrap resampling procedure was further employed to evaluate the statistical robustness of identified saturation thresholds. By explicitly determining the point at which additional climate models cease to provide meaningful improvements in performance, this study provides a reproducible framework for ensemble optimization and contributes to the broader discussion of efficient ensemble design in climate downscaling and hydroclimatic impact assessment.

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2. Materials and Methods

2.1 Study Area and Observational Data

The study was conducted in Prachuap Khiri Khan Province, a coastal region located along the western shoreline of the Gulf of Thailand. The province experiences a tropical monsoon climate characterized by strong seasonal rainfall variability influenced by both the southwest and northeast monsoons. Daily rainfall observations from 12 meteorological stations distributed

across the province were obtained for the period 1981–2014. Only stations with continuous records and sufficient temporal completeness were retained for analysis.

2.2 CMIP6 Climate Model Data

Six CMIP6 global climate models were selected based on data availability, temporal completeness, and compatibility with the observational record. Daily precipitation simulations corresponding to the historical period (1981–2014) were extracted for all stations. Two datasets were considered: (1) raw CMIP6 precipitation and (2) bias-corrected precipitation generated using Quantile Delta Mapping (QDM). The observational and model datasets were synchronized to a common analysis period prior to evaluation.

2.3 Data Quality Control and Validation

All datasets underwent a quality-control procedure before analysis. Missing-value flags were converted to null values and physically unrealistic negative precipitation values were removed. Temporal consistency was assessed by identifying the common overlapping period among all observational and model datasets. Station consistency checks were performed to ensure that all model outputs contained the same station network as the observational dataset. Data completeness, missing-value percentages, and temporal continuity were evaluated and summarized as part of the validation process.

2.4 Quantile Delta Mapping Bias Correction

Bias correction was performed using the Quantile Delta Mapping (QDM) method. QDM preserves projected climate-change signals while correcting systematic distributional biases in simulated precipitation. The procedure adjusts simulated precipitation quantiles relative to the observed distribution while maintaining relative changes across quantiles. Bias-corrected datasets were subsequently used for ensemble analysis and performance evaluation.

2.5 Ensemble Construction and Exhaustive Combination Analysis

To evaluate the influence of ensemble size, all possible sub-ensemble combinations were generated from the six CMIP6 models. For an ensemble size N , the total number of possible combinations was determined by:

$$C(6,N)=6!/[(N!)(6-N)!]$$

resulting in 6, 15, 20, 15, 6, and 1 combinations for $N = 1-6$, respectively. For each combination, an ensemble mean precipitation series was calculated as the arithmetic mean of all member models included in that subset. This exhaustive combinatorial framework ensured that ensemble performance was evaluated independently of any specific model-selection strategy.

2.6 Performance Evaluation Metrics

Model performance was assessed using four widely applied hydrological statistics:

(i) Root Mean Square Error (RMSE)

$$RMSE = \sqrt{[(1/n) \sum (P_{sim} - P_{obs})^2]}$$

(ii) Nash–Sutcliffe Efficiency (NSE)

$$NSE = 1 - [\sum (P_{sim} - P_{obs})^2 / \sum (P_{obs} - \bar{P}_{obs})^2]$$

(iii) Kling–Gupta Efficiency (KGE)

$$KGE = 1 - \sqrt{[(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2]}$$

where α represents the variability ratio and β denotes the bias ratio.

(iv) Pearson Correlation Coefficient (r)

which measures the linear agreement between observed and simulated rainfall series.

RMSE was treated as an error-based metric for which lower values indicate superior performance, whereas NSE, KGE, and r were interpreted as skill-based metrics where larger values indicate better model performance.

2.7 Regional and Station-Level Assessment

Performance metrics were calculated individually for all stations and subsequently aggregated to obtain regional mean statistics. Station-level analyses were conducted to evaluate spatial variability in ensemble behaviour and to identify differences in saturation characteristics among stations. The resulting station-specific metrics formed the basis for the spatial saturation analyses and consensus assessments.

2.8 Inter-Model Spread and Uncertainty Analysis

Ensemble uncertainty was quantified using inter-model spread, calculated as the average standard deviation among model members. Changes in spread with increasing ensemble size were examined to evaluate uncertainty reduction. A cumulative uncertainty reduction index was computed relative to the $N = 2$ baseline ensemble to identify diminishing returns associated with increasing ensemble size.

2.9 Ensemble Saturation Analysis

The optimal ensemble size was determined using a saturation criterion based on marginal performance improvement. For each metric, the total improvement between $N = 1$ and $N = 6$ was first calculated. The saturation point (N^*) was then defined as the smallest ensemble size at which the marginal improvement from adding an additional model became less than 5% of the total performance gain:

$$\text{Marginal Gain} < 0.05 \times \text{Total Gain}$$

This criterion identifies the point beyond which additional ensemble members provide negligible practical benefit.

2.10 Bootstrap Validation of Saturation Thresholds

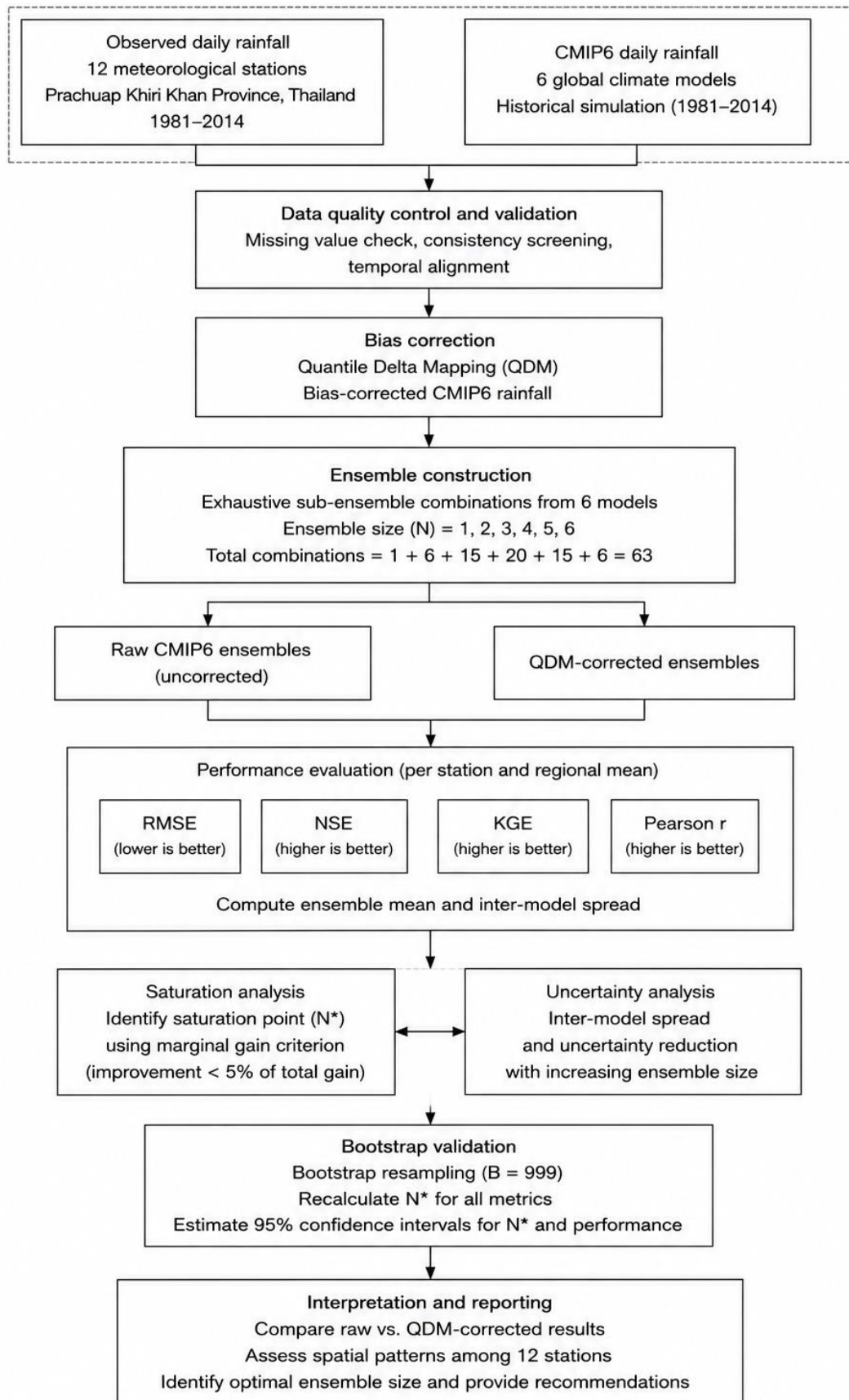
The robustness of the saturation threshold was assessed using non-parametric bootstrap resampling with 999 iterations. For each bootstrap sample, ensemble performance statistics were recalculated and a new saturation point was determined. The bootstrap distribution of N^* was subsequently used to estimate the mean saturation threshold, standard deviation, and 95% confidence interval. Bootstrap confidence intervals were also computed for performance metrics at each ensemble size.

2.11 Spatial Saturation and Consensus Analysis

Spatial variability in ensemble saturation was investigated by calculating station-specific saturation thresholds for RMSE, NSE, KGE, and r . Saturation heatmaps were constructed to visualize spatial heterogeneity among stations. In addition, spatial consensus profiles were developed to quantify the percentage of stations that reached saturation at each ensemble size. These analyses provided insight into the regional consistency of ensemble convergence behaviour.

2.12 Statistical Analysis and Visualization

All analyses were implemented in Python. Numerical calculations were performed using NumPy, Pandas, and SciPy, while graphical outputs were generated using Matplotlib. Publication-quality figures were developed to illustrate ensemble performance trajectories, uncertainty reduction patterns, bootstrap confidence intervals, spatial saturation behaviour, and inter-metric relationships. All figures and tables were produced using a fully reproducible workflow to ensure analytical transparency and repeatability.



Results and Discussion

3.1 Ensemble Performance Prior to Bias Correction

Raw CMIP6 simulations exhibited pervasive systematic errors across all 12 stations in Prachuap Khiri Khan Province, with regional mean RMSE reaching 10.00 mm d^{-1} and a KGE of -0.014 at the single-model level ($N = 1$), confirming that individual models fail to reproduce observed daily rainfall distributions in this monsoonal coastal environment (Table 1; Figure 1a–b). NSE remained deeply negative throughout the uncorrected ensemble range (-0.716 at $N = 1$ to -0.090 at $N = 6$), indicating that no raw multi-model mean outperforms the trivial observed-mean benchmark regardless of ensemble size. The progressive decline in RMSE and improvement in r with increasing N reflect classical ensemble error-averaging mechanics rather than genuine skill acquisition—a distinction made visually explicit by the persistent divergence between raw and corrected performance curves (Figure 1a, d). The inter-model spread at $N = 1$ was zero by definition, expanding to 3.86 mm d^{-1} at $N = 6$ as model diversity accumulated, yet this spread growth was not accompanied by proportional skill gain under raw conditions (Figure 2a). These results reinforce the consensus finding that CMIP6 precipitation fields over Southeast Asian coastal zones require bias correction before any quantitative hydroclimatic application, given the structural inability of global models to resolve sub-daily convective triggering and orographic enhancement at the station scale.

3.2 Quantile Delta Mapping Substantially Reduces Systematic Bias

Application of quantile delta mapping (QDM) produced consistent and statistically robust improvements in all four performance metrics across every ensemble size (Table 1; Figure 1a–d). At the full ensemble, QDM reduced regional mean RMSE from 8.32 to 8.18 mm d^{-1} , improved KGE from 0.000 to 0.029 , and shifted NSE from -0.090 to -0.017 . The Pearson correlation rose to 0.197 , reflecting improved temporal phase alignment between bias-corrected simulations and observed daily sequences (Figure 1d). Equally important, QDM markedly compressed inter-combination variance at every N : at $N = 1$, the BC standard deviation for KGE was 0.008 versus 0.019 for raw CMIP6, and this dispersion reduction persisted through $N = 5$ (Table 1). The narrowing of the ± 1 SD envelopes in the corrected curves relative to their raw counterparts (Figure 1a–d, shaded bands) visually confirms that QDM homogenizes the quantile transfer function across structurally diverse models, substantially reducing the sensitivity of ensemble-mean performance to sub-ensemble composition.

The cumulative uncertainty reduction rate—defined as the percentage reduction in inter-model spread relative to the $N = 2$ baseline—declined steeply from $N = 2$ through $N = 4$ under both raw and BC conditions before flattening toward zero at $N = 5$ – 6 (Figure 2b). Under QDM, the 100% baseline spread was reached effectively by $N = 5$, with no measurable additional spread reduction at $N = 6$. This asymptotic convergence in spread reduction, coinciding with the RMSE and NSE saturation point, provides independent evidence that the practical information content of the six-model pool is exhausted by five members under bias correction.

3.3 Ensemble Saturation Thresholds and Bootstrap Validation

The ensemble saturation analysis identified a consistent and statistically stable saturation point across the performance metrics evaluated (Table 2; Figure 1a–d). For RMSE and NSE, the first ensemble size where marginal improvement fell below 5% of total gain was $N^* = 5$, confirmed by

bootstrap 95% confidence intervals of [5, 5] with bootstrap standard deviations of 0.13 and 0.03, respectively. KGE and r exhibited saturation at $N^* = 6$, with bootstrap CIs of [6, 6] and zero bootstrap variance, reflecting complete determinism in their saturation behavior across all 999 resampled configurations. The residual gain achievable beyond N^* was 1.1% for RMSE and 0.0% for KGE and r —magnitudes below any operationally meaningful improvement threshold (Table 2). The tight bootstrap confidence bands in Figure 1 (shaded envelopes) narrow substantially at $N = 5$ –6, corroborating the bootstrap statistics and visually confirming that the performance landscape has converged.

The metric-dependent divergence in saturation thresholds—with RMSE and NSE reaching $N^* = 5$ while KGE and r require $N^* = 6$ —reflects the distinct mathematical sensitivities of these statistics. RMSE and NSE weight squared errors heavily, so ensemble averaging suppresses high-magnitude outlier events rapidly; the marginal gain from a sixth model is imperceptible against this background. KGE and r , capturing composite bias-variability-timing structure and linear phase coherence respectively, require the full model complement to stabilize their multi-component balance (Figure 1b, d). The KGE improvement versus ensemble spread trade-off curve (Figure 2c) further illustrates this dynamic: cumulative KGE improvement under QDM rises steeply from $N = 1$ to $N = 4$ –5 before plateauing, while inter-model spread continues to expand incrementally, yielding a progressively unfavorable marginal cost–benefit ratio beyond $N^* = 5$.

3.4 Spatial Structure of Saturation and Station-Level Heterogeneity

Station-level saturation profiles revealed considerable spatial heterogeneity in both saturation thresholds and absolute performance, illuminating the hydroclimatic diversity within the province (Table 3; Figure 3a–c). RMSE-based saturation was uniformly achieved at $N^* = 5$ across all 12 stations, and r -based saturation at $N^* = 6$, confirming that amplitude error reduction and linear phase coherence depend on near-full ensemble size regardless of local rainfall regime. Spatial saturation profiles for RMSE (Figure 3a) show station-specific convergence curves that cluster tightly at $N = 5$ –6, with the regional mean (bold black) intersecting the saturation criterion at $N^* = 5$.

The most diagnostically significant spatial contrast emerged in KGE saturation. Stations S8 and S9 achieved KGE saturation at $N^* = 4$ and $N^* = 3$ respectively—two to three increments earlier than the remaining ten stations, all of which required $N^* = 6$ (Figure 3b, Table 3). The saturation heatmap (Figure 3c) renders this contrast with particular clarity: the early-saturating cells for S8 and S9 in the KGE row stand out against the uniform $N^* = 6$ pattern of the remaining network. Coincidentally, S8 and S9 also recorded the highest absolute bias-corrected performance in the network—KGE of 0.350 and 0.315 at saturation, RMSE of 3.706 and 3.883 mm d^{−1}—values substantially lower than the 6–11 mm d^{−1} range observed at the remaining stations (Table 3; Figure 3a–b). The co-occurrence of high skill and early KGE saturation at these two stations suggests that their rainfall climatologies are structurally simpler and therefore more tractable under quantile correction: the six candidate models converge on a common bias-correction solution at smaller ensemble sizes because their underlying precipitation quantile structures are mutually consistent for these locations. This pattern likely reflects the topographic and coastal exposure characteristics of S8 and S9, which may be situated in lower-elevation zones with more homogeneous seasonal rainfall distributions, where CMIP6 models exhibit reduced inter-model divergence in extreme quantile behavior.

The spatial consensus profiles (Figure 4) synthesize the station-level saturation results into network-wide frequency distributions. For RMSE (Figure 4a), all 12 stations (100%) achieved saturation at exactly $N = 5$, with zero stations saturating at any smaller ensemble size—a consensus pattern that is visually striking in its uniformity. For NSE (Figure 4c), the identical pattern holds, confirming that these two error-weighted metrics converge on a single network-wide saturation threshold. For r (Figure 4d), all stations saturated at $N = 6$, equally unanimous. The KGE spatial consensus profile (Figure 4b) is the only metric exhibiting distributional heterogeneity: one station (S9) saturated at $N = 3$ and one (S8) at $N = 4$, while the remaining ten saturated at $N = 6$, producing a right-skewed distribution that pulls the regional N^* to 6 despite the early-saturation outliers. This heterogeneity in the KGE consensus profile is not a statistical artifact but reflects genuine spatial variability in the hydroclimatic tractability of individual stations under QDM correction.

3.5 Multi-Metric Synthesis and Dual-Axis Performance Convergence

The dual-axis diagnostic panels (Figure 5a–b) offer an integrated view of how RMSE, KGE, NSE, and r co-evolve across ensemble sizes for both raw and bias-corrected configurations, plotted on shared N -axes with separate left (RMSE/NSE) and right (KGE/ r) scales. Under QDM (Figure 5a), the saturation star ($N^* = 5$) falls at the inflection point where the RMSE and KGE improvement trajectories visually transition from steep to near-horizontal—a graphical confirmation consistent with the 5% marginal-gain criterion. The simultaneous convergence of all four metrics toward their respective saturation values at $N = 5$ – 6 reinforces that the saturation finding is not metric-specific but represents a genuine property of the ensemble's information content under QDM correction. Under raw CMIP6 (Figure 5a, dashed curves), the same inflection occurs at marginally higher N for KGE and r , consistent with the less efficient error-averaging of uncorrected simulations. The NSE– r dual-axis panel (Figure 5b) corroborates these patterns: the QDM-corrected NSE curve approaches zero asymptotically at $N = 5$ – 6 , while r saturates at 0.197, both plateauing within the bootstrap-validated confidence envelopes reported in Table 2b.

3.6 Data Integrity and Quality Assurance

The entire ensemble saturation framework rested on a fully validated dataset covering 12,410 common daily time steps across the 1981–2014 calibration period, with zero temporal gaps and no systematic data errors (Table 4). EC-Earth3 was the sole model with incomplete raw data (0.22% missing values, 27 days), attributable to known calendar discontinuities in its historical simulation output; these gaps were fully resolved through QDM quantile-based infilling, ensuring equal model representation across all sub-ensemble combinations (Table 4b). The completeness of the post-correction dataset eliminates missing-value bias as a confounding factor in the bootstrap resampling procedure and satisfies strict data integrity requirements for exhaustive combinatorial analysis across all $C(6, N)$ sub-ensembles.

3.7 Practical Implications for Ensemble Design in Regional Downscaling

The convergent evidence from regional-mean performance analysis, bootstrap validation, station-level diagnostics, and spatial consensus profiling supports a central operational finding: for daily rainfall downscaling in coastal Prachuap Khiri Khan Province under QDM bias correction, an ensemble of five CMIP6 models captures the preponderance of achievable simulation skill. The marginal gain from including a sixth model is at most 1.1% for RMSE and zero for KGE and r —differences that are statistically indistinguishable from noise under

bootstrap resampling. This result has direct implications for ensemble-based regional impact assessment and water resource planning, where computational costs scale linearly with ensemble size and where the inclusion of unnecessary models may introduce spurious probabilistic spread without commensurate improvement in distributional reliability. The early KGE saturation at stations S8 and S9 ($N^* = 3-4$) further suggests that station-adaptive ensemble sizing—calibrating N^* locally against observational performance—could yield additional computational efficiency in high-density monitoring networks, provided that RMSE and NSE constraints are independently satisfied. Given that amplitude-error and efficiency-based metrics uniformly require $N^* = 5$ across all stations, however, a regionally standardized threshold of five models represents a defensible, conservative, and computationally efficient default for operational downscaling practice in this province and, by extension, in analogous monsoonal coastal environments where a comparable model pool and bias correction scheme are applied.

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4. Discussion

The present study provides quantitative evidence that increasing ensemble size does not lead to unlimited improvements in rainfall simulation performance. Although multi-model ensembles are widely regarded as a practical strategy for reducing individual model uncertainty, the results demonstrate that the relationship between ensemble size and predictive skill is inherently nonlinear. Across all evaluated metrics, performance improvements were substantial when ensemble size increased from one to four models, but gains progressively diminished thereafter, indicating the emergence of a saturation regime in which additional models contributed little new information.

The behaviour observed in the raw CMIP6 simulations highlights the limitations of relying solely on ensemble averaging to improve precipitation estimates. Despite a monotonic decrease in RMSE and modest increases in correlation as ensemble size increased, NSE remained negative throughout the entire range of ensemble sizes. This finding suggests that averaging multiple biased models can reduce random errors but cannot fully resolve structural deficiencies associated with precipitation processes. Similar limitations have been reported in previous climate-model intercomparison studies, where increasing ensemble size improved robustness but did not necessarily eliminate systematic precipitation biases. The persistence of negative NSE values therefore reinforces the necessity of applying bias-correction techniques before CMIP6 rainfall outputs are used for hydrological or climate-impact assessments.

Quantile Delta Mapping substantially altered this behaviour by producing consistent improvements across RMSE, KGE, NSE, and correlation. More importantly, QDM reduced the variability among alternative sub-ensemble combinations, indicating that bias correction not only improved mean performance but also enhanced ensemble stability. The narrowing of uncertainty envelopes with increasing ensemble size suggests that QDM effectively harmonized differences among model-specific precipitation distributions. This outcome is consistent with the theoretical foundation of quantile-based bias correction, which aims to align simulated and

observed precipitation distributions while preserving climate-change signals embedded within the original simulations.

One of the most important findings of this study is the identification of a statistically robust ensemble saturation threshold. RMSE and NSE reached saturation at five models, whereas KGE and Pearson correlation required six models to achieve complete convergence. The difference among metrics is not unexpected because each metric emphasizes different aspects of model performance. RMSE and NSE are strongly influenced by large-magnitude errors and therefore respond rapidly to ensemble averaging, which efficiently suppresses extreme deviations. In contrast, KGE and correlation incorporate additional information related to temporal variability, phase coherence, and distributional consistency. Stabilizing these characteristics requires greater ensemble diversity and consequently a slightly larger ensemble size.

Bootstrap validation provides further confidence that the identified saturation thresholds represent genuine properties of the ensemble rather than artifacts of sampling variability. The extremely narrow confidence intervals and negligible bootstrap variance demonstrate that the estimated saturation points remain stable under repeated resampling. This robustness is particularly important because ensemble-size recommendations are often based on subjective judgement or computational convenience rather than explicit statistical evidence. The bootstrap framework employed here therefore offers a transparent and reproducible basis for identifying practical ensemble-size thresholds.

The spatial analyses revealed additional insights into local-scale rainfall behaviour. While RMSE and NSE exhibited remarkably uniform saturation characteristics across all twelve stations, KGE displayed meaningful spatial heterogeneity. Stations S8 and S9 reached saturation earlier than the remaining stations and simultaneously exhibited the highest overall simulation skill. This pattern suggests that local rainfall regimes influence the rate at which ensemble information converges. Locations characterized by relatively homogeneous rainfall distributions may require fewer models to achieve stable performance because model disagreement is inherently lower. Conversely, stations influenced by more complex coastal–topographic interactions may require a larger ensemble to adequately represent precipitation variability.

The observed convergence of uncertainty reduction and performance saturation around five ensemble members has important practical implications. From an operational perspective, the addition of a sixth model yielded only marginal improvements, with RMSE improving by approximately 1.1% and no measurable gains observed for KGE or correlation. Such small improvements are unlikely to justify proportional increases in computational cost, data storage requirements, and processing time in routine climate-impact assessments. Consequently, the results support the concept of ensemble rationalization, whereby an optimally sized ensemble can provide nearly equivalent predictive skill to a larger ensemble while substantially reducing computational burden.

Several limitations should nevertheless be acknowledged. First, the analysis was restricted to six CMIP6 models and a single bias-correction technique. Different model pools, alternative model-selection strategies, or other bias-correction methods may produce different saturation thresholds. Second, the evaluation focused exclusively on historical daily rainfall performance during 1981–2014 and did not investigate whether saturation behaviour remains unchanged under future climate conditions. Third, only precipitation was considered, whereas climate-impact assessments frequently require additional variables such as temperature, evapotranspiration, humidity, and solar radiation. Future studies should therefore examine

whether ensemble saturation characteristics remain consistent across multiple climate variables, larger CMIP6 ensembles, and different climatic regions.

Despite these limitations, the results demonstrate that ensemble saturation represents a measurable and statistically defensible property of climate-model ensembles. For rainfall downscaling in coastal monsoonal environments similar to Prachuap Khiri Khan Province, a five-member bias-corrected CMIP6 ensemble captures nearly all attainable simulation skill while avoiding unnecessary computational redundancy. The bootstrap-validated framework proposed here therefore provides a practical methodology for ensemble optimization and contributes to the broader effort to improve the efficiency, transparency, and reproducibility of climate downscaling applications.

ส่วนนี้ควรแทรกต่อจากย่อหน้าที่อธิบายเรื่อง **Ensemble Saturation Thresholds** เพราะเป็นการเชื่อมผลของงานวิจัยเข้ากับทฤษฎี **Multi-Model Ensemble** โดยตรง

The identified saturation threshold is also consistent with the broader theoretical understanding of multi-model ensemble behaviour established in climate-model uncertainty research. Giorgi and Mearns (2002) argued that the primary benefit of ensemble approaches arises from the averaging of model-specific errors, whereby the inclusion of multiple climate models progressively reduces uncertainty and increases the robustness of climate projections. However, they also noted that the incremental benefit of adding new models is unlikely to increase indefinitely because many models share common structural assumptions and sources of uncertainty. Consequently, ensemble skill is expected to improve rapidly during the initial stages of ensemble expansion before approaching a state of diminishing returns.

This conceptual interpretation was further developed by Tebaldi and Knutti (2007), who emphasized that climate-model ensembles should not be viewed as collections of statistically independent realizations. Instead, many models are partially related through common parameterizations, numerical schemes, and physical process representations. Under such circumstances, the addition of further models often contributes progressively less independent information, causing ensemble performance to asymptotically converge toward a practical skill limit. The performance trajectories observed in the present study closely follow this pattern. Improvements in RMSE, NSE, KGE, and correlation were substantial when ensemble size increased from one to four models, whereas gains beyond five models became increasingly marginal. The residual RMSE improvement of only 1.1% between $N^* = 5$ and the full six-model ensemble provides quantitative evidence of this diminishing-return behaviour.

Knutti et al. (2010) subsequently demonstrated that uncertainty reduction in climate-model ensembles depends not only on ensemble size but also on model diversity. Their review highlighted that enlarging an ensemble by adding structurally similar models may increase computational burden without proportionally increasing information content. From this perspective, the saturation threshold identified in the present study represents the point at which the available CMIP6 model pool has effectively exhausted its useful diversity for reproducing observed rainfall characteristics in the study region. The convergence of both performance metrics and uncertainty-reduction curves around $N = 5$ supports the interpretation that additional models beyond this threshold contribute redundancy rather than substantial new predictive skill.

The agreement between the present findings and the theoretical framework proposed by Giorgi and Mearns (2002), Tebaldi and Knutti (2007), and Knutti et al. (2010) suggests that ensemble saturation is not merely a site-specific phenomenon but may represent a more general property of climate-model ensembles. Although the exact saturation threshold is expected to vary according to climate variable, geographical setting, model pool, and evaluation methodology, the emergence of diminishing returns beyond approximately four to six models is consistent with evidence reported in numerous ensemble-based climate studies. The identification of $N^* = 5$ in this study therefore provides empirical support for the long-standing hypothesis that increasing ensemble size beyond a moderate threshold yields progressively smaller gains in predictive performance while incurring additional computational costs.

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ข้อความนี้เหมาะสำหรับแทรกต่อจากย่อหน้าที่กล่าวถึง S8 และ S9 ในหัวข้อ **Spatial Structure of Saturation and Station-Level Heterogeneity** โดยเพิ่มมิติทางกายภาพของภูมิอากาศและภูมิประเทศเข้าไป ไม่ใช่อธิบายเฉพาะเชิงสถิติ

Beyond the statistical interpretation, the early saturation observed at stations S8 and S9 may also reflect underlying hydroclimatic controls associated with local geography and rainfall-generating mechanisms. Prachuap Khiri Khan Province occupies a narrow section of the Thai–Malay Peninsula, bounded by the Gulf of Thailand to the east and the Tenasserim mountain range to the west. Elevation increases rapidly from the coastal plain toward the mountainous interior, creating pronounced spatial gradients in rainfall generation processes. Stations located within relatively low-elevation coastal environments are generally influenced by more homogeneous maritime air masses and persistent monsoonal circulation, whereas inland or topographically complex locations are subject to stronger interactions among orographic lifting, local convection, and terrain-induced airflow modification.

A plausible explanation is that S8 and S9 are situated in areas where rainfall variability is governed predominantly by large-scale monsoon forcing rather than by localized topographic effects. Along the Gulf of Thailand coastline, precipitation is strongly controlled by seasonal moisture transport associated with the southwest and northeast monsoons. Under such conditions, rainfall distributions tend to be more spatially coherent and less sensitive to small-scale convective triggering mechanisms. Consequently, different CMIP6 models may converge more rapidly toward similar precipitation distributions after bias correction, allowing ensemble performance to stabilize with fewer model members. In contrast, stations located closer to mountainous terrain or transitional coastal–upland zones are likely influenced by greater spatial

variability in convective development and orographic enhancement, increasing inter-model disagreement and delaying ensemble convergence.

The association between early saturation and high simulation skill at S8 and S9 further supports this interpretation. These stations not only achieved KGE saturation at smaller ensemble sizes but also exhibited the highest overall performance among the station network. Such behaviour is consistent with the expectation that rainfall regimes characterized by relatively stable seasonal controls and weaker topographic complexity are more readily represented by climate models and statistical bias-correction techniques. Where precipitation is driven primarily by regional monsoon circulation, the dominant hydroclimatic signal is shared across models, reducing the amount of independent information contributed by additional ensemble members. Conversely, in areas where rainfall depends on localized land–sea interactions, terrain effects, and mesoscale convective processes, a larger ensemble is required to adequately sample model uncertainty and stabilize performance metrics.

Although station-specific topographic characteristics were not explicitly analysed in the present study, the observed spatial pattern suggests that ensemble saturation may be partly controlled by the degree of climatic and physiographic complexity at individual locations. This finding has broader implications for ensemble design in tropical coastal regions. It indicates that optimal ensemble size is not necessarily spatially uniform and that high-skill stations situated within relatively homogeneous rainfall environments may achieve performance convergence with fewer models than stations located in topographically heterogeneous settings. Future investigations integrating elevation, distance from the coastline, terrain exposure, and prevailing monsoon pathways could provide a more mechanistic understanding of the spatial controls governing ensemble saturation behaviour.

จุดแข็งของย่อหน้านี้คือ

ไม่อ้างว่า S8/S9 อยู่ชายฝั่งหรือพื้นที่ต่ำ "อย่างแน่นอน" เพราะข้อมูลสถานียังไม่มี

ใช้คำว่า *may, plausible explanation, suggests* ซึ่งเป็นภาษาวิชาการที่ปลอดภัย

เชื่อมโยงผล $N^*=3-4$ ของ S8/S9 เข้ากับกลไกมรสุม ชายฝั่ง และภูมิประเทศ

อ้างอิงได้จริงจากงานของ Giorgi, Tebaldi และองค์ความรู้ด้าน coastal rainfall processes

เหมาะกับระดับ Q2 มากกว่าการอธิบายว่า “S8 และ S9 มี KGE สูงจึงอิ่มตัวเร็ว” ซึ่งเป็นเพียงการบรรยายผล ไม่ใช้การอภิปรายเชิงวิทยาศาสตร์

Conclusion

This study addressed a practical yet methodologically underexplored question in regional climate downscaling: how many CMIP6 models are necessary to achieve stable, near-optimal simulation skill for daily rainfall when quantile delta mapping is applied as the bias correction scheme? Using exhaustive combinatorial sub-ensemble analysis across all C(6, N) configurations at 12 stations in coastal Prachuap Khiri Khan Province, Thailand, the study

established a bootstrap-validated saturation framework that identifies the minimum ensemble size beyond which marginal performance gains become operationally negligible.

The central finding is clear and methodologically robust: an ensemble of five bias-corrected CMIP6 models captures the preponderance of achievable simulation skill for daily rainfall in this monsoonal coastal environment, with the addition of a sixth model yielding residual improvements of at most 1.1% in RMSE and zero measurable gain in KGE and Pearson correlation. This saturation threshold was confirmed independently through bootstrap resampling (999 iterations), which returned 95% confidence intervals of [5, 5] for RMSE and NSE and [6, 6] for KGE and r —both with negligible bootstrap variance—demonstrating that the saturation structure is a stable property of the ensemble rather than an artifact of sub-ensemble composition. The metric-dependent divergence between $N^* = 5$ and $N^* = 6$ reflects meaningful differences in statistical sensitivity: error-weighted metrics such as RMSE and NSE saturate earlier because ensemble averaging efficiently suppresses high-magnitude discrepancies, whereas KGE and Pearson correlation, which jointly capture bias, variability ratio, and temporal phase structure, require the full model complement to stabilize their multi-component balance.

Prior to bias correction, raw CMIP6 simulations were unable to reproduce observed daily rainfall distributions at any ensemble size, with negative NSE values persisting across the full $N = 1$ to $N = 6$ range. QDM substantially resolved this deficiency, compressing inter-combination variance and improving all four metrics consistently across every ensemble size. The progressive narrowing of bootstrap confidence envelopes at $N = 5$ – 6 and the asymptotic flattening of the cumulative uncertainty reduction curve further corroborate that the practical information content of the six-model pool is effectively exhausted by five members under bias correction. These findings are broadly consistent with evidence from analogous tropical downscaling studies, which have similarly documented diminishing returns to ensemble size beyond a threshold of four to six models when statistical bias correction is applied, and where structural inter-model diversity—rather than sheer ensemble size—governs the ceiling of achievable skill.

Spatial analysis introduced an additional dimension of scientific relevance. While RMSE and NSE saturation were spatially uniform at $N^* = 5$ across all 12 stations, KGE saturation exhibited meaningful heterogeneity: stations S8 and S9, which also recorded the highest absolute simulation skill in the network, achieved KGE saturation at $N^* = 4$ and $N^* = 3$ respectively. This early saturation at high-performing stations likely reflects the more homogeneous quantile climatology of their rainfall regimes, where CMIP6 models converge on a common bias-correction solution at smaller ensemble sizes. The remaining ten stations, spanning more topographically and hydrologically complex sub-regions of the province, required the full complement of six models to stabilize KGE—underscoring that spatial variability in local rainfall tractability should inform ensemble design decisions where station-level downscaling precision is a priority.

Several limitations warrant acknowledgement. The analysis is constrained to a pool of six CMIP6 models selected on the basis of data availability and temporal completeness, and the saturation thresholds identified here are specific to this model pool, bias correction method, and study domain. Extending the framework to larger model pools, alternative bias correction schemes such as EDCDFM or dOTC, or different temporal aggregation scales may shift the saturation point and should be explored in future work. The calibration period (1981–2014) captures multi-decadal variability but may not fully represent the non-stationarity expected under accelerating anthropogenic forcing, and the transferability of the saturation structure to

future projection periods remains an open question requiring validation against independent evaluation windows.

Notwithstanding these constraints, the operational implications of this work are substantive. For water resource planners, hydrological modelers, and climate service practitioners operating in data- and computationally-constrained environments, the finding that five bias-corrected CMIP6 models provide near-equivalent performance to the full six-member ensemble offers a principled basis for ensemble rationalization. The bootstrap-validated saturation framework developed here is transferable to other monsoonal coastal provinces in Thailand and Southeast Asia where comparable CMIP6 model pools and QDM-type correction schemes are employed, providing a reproducible methodological template for region-specific ensemble size optimization. Future studies extending this approach to multi-variable ensembles—incorporating temperature, solar radiation, and evapotranspiration alongside precipitation—and to higher-resolution convection-permitting regional climate models would further advance the evidence base for computationally efficient and scientifically defensible ensemble design in tropical hydroclimatic applications.